YeshwantraoChavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)
Accredited 'A++' Grade by NAAC
Hingna Road, Wanadongri, Nagpur-441110

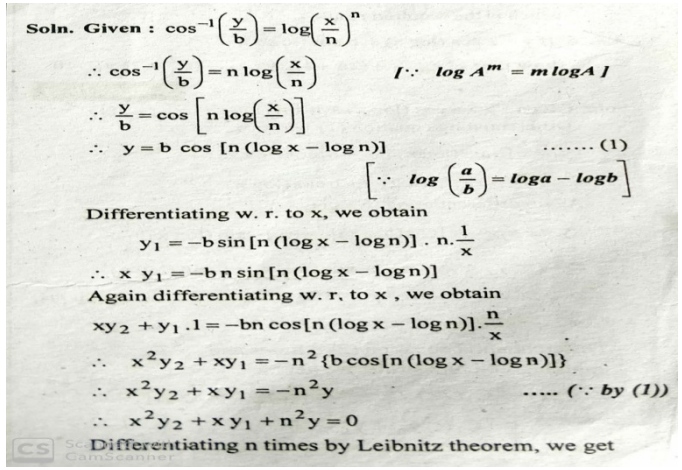
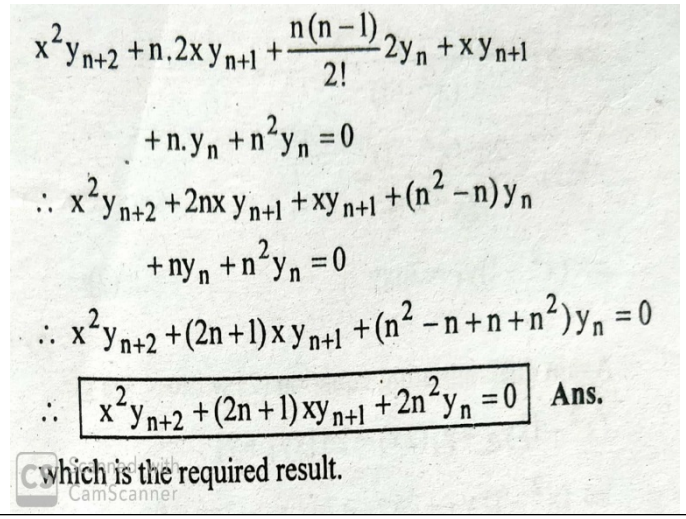
Department of Applied Mathematics & Humanities

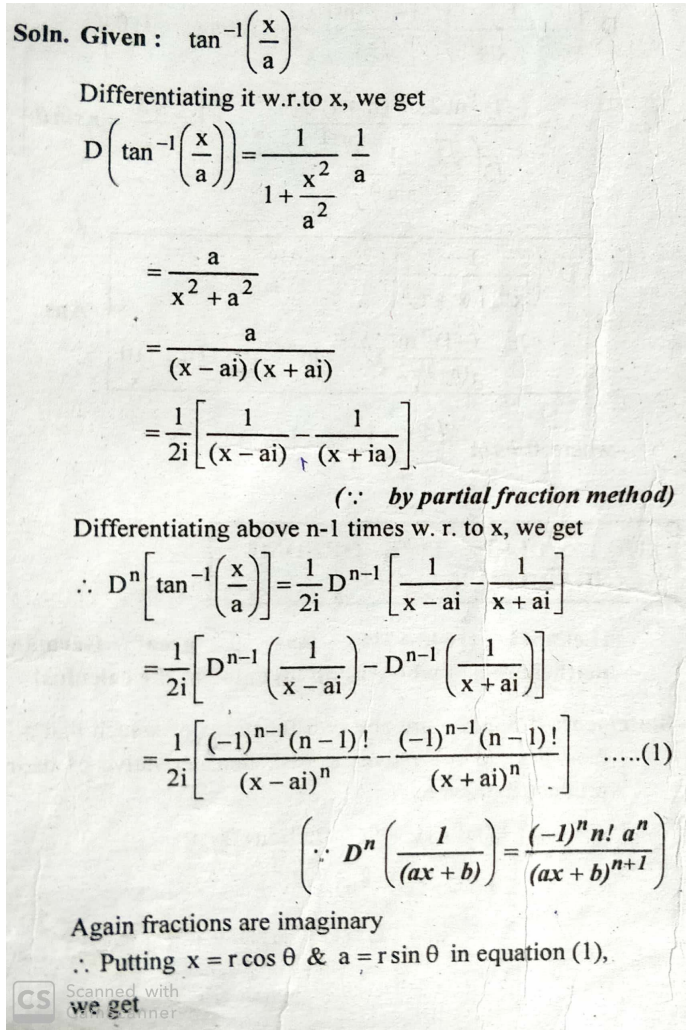
Electronic Question Answer Bank

Course Name: Calculus and Vector (C & V)

Course Code: 23GE1101

UNIT-I : DIFFERNTIAL CALCULUS

Sr.No.	Unit No.	Text of Question	Time required in Minutes	Marks allotted	lengthiness	Difficulty level	CO Mapping	Type Descriptive / Numerical
1.	Unit 1	If then show that  **Soln. Given :** $\cos^{-1}\left(\frac{y}{b}\right)=\log\left(\frac{x}{n}\right)^n$ $\therefore \cos^{-1}\left(\frac{y}{b}\right)=n\log\left(\frac{x}{n}\right)$ $[\because \log A^m = m\log A]$ $\therefore \frac{y}{b}=\cos\left[n\log\left(\frac{x}{n}\right)\right]$ $\therefore y=b\cos[n(\log x-\log n)]$ ..... (1) $\left[\because \log\left(\frac{a}{b}\right)=\log a-\log b\right]$ Differentiating w. r. to x, we obtain $y_1=-b\sin[n(\log x-\log n)]\cdot n.\frac{1}{x}$ $\therefore x\,y_1=-bn\sin[n(\log x-\log n)]$ Again differentiating w. r. to x , we obtain $xy_2+y_1.1=-bn\cos[n(\log x-\log n)].\frac{n}{x}$ $\therefore x^2y_2+xy_1=-n^2\{b\cos[n(\log x-\log n)]\}$ $\therefore x^2y_2+x\,y_1=-n^2y$ ..... $(\because$ by (1)) $\therefore x^2y_2+x\,y_1+n^2y=0$ Differentiating n times by Leibnitz theorem, we get  $x^2y_{n+2}+n.2x\,y_{n+1}+\frac{n(n-1)}{2!}2y_n+x\,y_{n+1}$ $+n.y_n+n^2y_n=0$ $\therefore x^2y_{n+2}+2nx\,y_{n+1}+xy_{n+1}+(n^2-n)y_n$ $+ny_n+n^2y_n=0$ $\therefore x^2y_{n+2}+(2n+1)x\,y_{n+1}+(n^2-n+n+n^2)y_n=0$ $\therefore \boxed{x^2y_{n+2}+(2n+1)xy_{n+1}+2n^2y_n=0}$ **Ans.** which is the required result.	15	5	Medium	Easy	CO1	Descriptive

	Unit 1	Find the derivative of 	10	5	Medium	Easy	CO1	Descriptive

Soln. Given : $\tan^{-1}\left(\frac{x}{a}\right)$

Differentiating it w.r.to x, we get

$$D\left(\tan^{-1}\left(\frac{x}{a}\right)\right)=\frac{1}{1+\frac{x^2}{a^2}}\,\frac{1}{a}$$

$$=\frac{a}{x^2+a^2}$$

$$=\frac{a}{(x-ai)(x+ai)}$$

$$=\frac{1}{2i}\left[\frac{1}{(x-ai)}-\frac{1}{(x+ia)}\right]$$

(∵ by partial fraction method)

Differentiating above n-1 times w. r. to x, we get

$$\therefore\ D^n\left[\tan^{-1}\left(\frac{x}{a}\right)\right]=\frac{1}{2i}D^{n-1}\left[\frac{1}{x-ai}-\frac{1}{x+ai}\right]$$

$$=\frac{1}{2i}\left[D^{n-1}\left(\frac{1}{x-ai}\right)-D^{n-1}\left(\frac{1}{x+ai}\right)\right]$$

$$=\frac{1}{2i}\left[\frac{(-1)^{n-1}(n-1)!}{(x-ai)^n}-\frac{(-1)^{n-1}(n-1)!}{(x+ai)^n}\right]\quad\ldots..(1)$$

$$\left(\because\ D^n\left(\frac{1}{(ax+b)}\right)=\frac{(-1)^n n!\,a^n}{(ax+b)^{n+1}}\right)$$

Again fractions are imaginary

∴ Putting $x=r\cos\theta$ & $a=r\sin\theta$ in equation (1),

we get

CS Scanned with CamScanner

		$$D^n\left[\tan^{-1}\left(\frac{x}{a}\right)\right] = \frac{1}{2i}\left[\frac{(-1)^{n-1}(n-1)!}{(r\cos\theta - ir\sin\theta)^n} - \frac{(-1)^{n-1}(n-1)!}{(r\cos\theta + ir\sin\theta)^n}\right]$$ $$= \frac{(-1)^{n-1}(n-1)!}{2i\,r^n}\left[\frac{1}{(\cos\theta - i\sin\theta)^n} - \frac{1}{(\cos\theta + i\sin\theta)^n}\right]$$ $$= \frac{(-1)^{n-1}(n-1)!}{2i\,r^n}\left[(\cos\theta - i\sin\theta)^{-n} - (\cos\theta + i\sin\theta)^{-n}\right]$$ $$= \frac{(-1)^{n-1}(n-1)!}{2i\,r^n}\left[(\cos n\theta + i\sin n\theta) - (\cos n\theta - i\sin n\theta)\right]$$ $$\left[\because (\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta \;\&\; \cos(-\theta) = \cos\theta,\ \sin(-\theta) = \sin\theta\right]$$ $$= \frac{(-1)^{n-1}(n-1)!}{2i\,r^n}[2i\sin n\theta]$$ $$= \frac{(-1)^{n-1}(n-1)!\sin n\theta}{r^n}$$ $$= \frac{(-1)^{n-1}(n-1)!\sin n\theta}{\left(\frac{a}{\sin\theta}\right)^n} \quad [\because a = r\sin\theta]$$ $$\therefore \boxed{D^n\left(\tan^{-1}\left(\frac{x}{a}\right)\right) = \frac{(-1)^{n-1}(n-1)!\sin^n\theta\,\sin n\theta}{a^n}}$$ Ans. where, $\theta = \tan^{-1}\left(\frac{a}{x}\right)$. Scanned with CamScanner					
3.	Unit 1	If then show that	10		Medium	Easy	CO1	Descriptive

		Soln. Given : $y = a\cos(\log x) + b\sin(\log x)$ ....(1) Differentiating equation (1) w.r.to. x, we get $y_1 = a[-\sin(\log x)].\frac{1}{x} + b[\cos(\log x)]\frac{1}{x}$ $\therefore\ xy_1 = -a\sin(\log x) + b\cos(\log x)$ Again differentiating, we get $xy_2 + y_1 = -a[\cos(\log x)]\frac{1}{x} + b[-\sin(\log x)]\frac{1}{x}$ $\Rightarrow\ x^2y_2 + xy_1 = -\{a\cos(\log x) + b\sin(\log x)\}$ $\Rightarrow\ x^2y_2 + xy_1 = -y$ *(∵ by (1))* $\Rightarrow\ x^2y_2 + xy_1 + y = 0$ ... (2) Differentiating equation (2) n times and applying Leibnitz's rule, we get $x^2y_{n+2} + {}^nC_1(2x)y_{n+1} + {}^nC_2(2).y_n + xy_{n+1} + {}^nC_1(1)y_n + y_n = 0$ Scanned with CamScanner $\Rightarrow x^2y_{n+2} + 2nxy_{n+1} + n(n-1)y_n + xy_{n+1} + ny_n + y_n = 0$ $\left[\because\ {}^nC_r = \frac{n(n-1)....(n-r+1)}{r!}\right]$ $\therefore\ \boxed{x^2y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0}$ **Ans.** Hence the result.		5				
4.	Unit 1	If prove that	10		Medium	Easy	CO1	Descriptive

		Soln. Given : $y = \cos(m \sin^{-1} x)$ ... (1) Differentiating w. r. to x, we get $y_1 = \frac{-m}{\sqrt{1-x^2}} \sin(m \sin^{-1} x)$ $\therefore \sqrt{1-x^2}\, y_1 = -m \sin(m \sin^{-1} x)$ On squaring both sides, we get $(1-x^2)y_1^2 = m^2 \sin^2(m \sin^{-1} x)$ $= m^2[1 - \cos^2(m \sin^{-1} x)]$ $\therefore (1-x^2) y_1^2 = m^2(1-y^2)$ (1) ($\because$ *by(1)*) Differentiating again w.r.to x, we get $(1-x^2)2y_1y_2 - 2xy_1^2 = m^2(-2yy_1)$ $\Rightarrow ((1-x^2)y_2 - xy_1 + m^2y)2y_1 = 0$ $\therefore (1-x^2)y_2 - xy_1 + m^2y = 0$ ....(3) Differentiating (2) n times by using Leibnitz's rule, we get $(1-x^2)y_{n+2} + n(-2x)y_{n+1}$ (2) $+ \frac{n(n-1)}{2}(-2)y_n - xy_{n+1} - ny_n + m^2y_n = 0$ $\therefore \boxed{(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2-n^2)y_n = 0}$ **Ans.** which is the required result.		5				
5.	Unit 1	If prove that	10		Medium	Easy	CO1	Descriptive

		Ans. Given : $y = e^{\tan^{-1}x}$ ...(1) Differentiating with respect to x, we get $y_1 = e^{\tan^{-1}x} \cdot \frac{1}{1+x^2}$ $\therefore (1+x^2)y_1 = e^{\tan^{-1}x} = y$ ...(2) [$\because$ *by (1)*] $\therefore (1+x^2)y_1 - y = 0$ Differentiating again, we get $(1+x^2)y_2 + 2xy_1 - y_1 = 0$ ...(3) Differentiating (3) w.r. to x, n times using Leibnitz's rule $D^n(uv) = u_n v + nu_{n-1}v_1 + \frac{n(n-1)}{2!}u_{n-2}v_2 + \ldots\ldots + v_n u,$ we get $D^n[(1+x^2)y_2] + D^n(2xy_1) - D^n y_1 = 0$ $\Rightarrow (1+x^2)y_{n+2} + n2xy_{n+1} + \frac{n(n-1)}{2}2y_n$ $+2(xy_{n+1} + ny_n) - y_{n+1} = 0$ $\therefore \boxed{(1+x^2)y_{n+2} + [2(n+1)x - 1]y_{n+1} + n(n+1)y_n = 0}$ **Ans** which is the required result.		5				
6.	Unit 1	If prove that	10		Medium	Easy	CO1	Descriptive

		Soln. Given : $y = \sin^{-1} x$ Differentiating w.r.to x, we get $y_1 = \frac{1}{\sqrt{1-x^2}}$ (½) $\Rightarrow \sqrt{1-x^2}\, y_1 = 1$ On squaring both sides $\Rightarrow (1-x^2) y_1^2 = 1$ Again differentiating w.r.to x, we get $(1-x^2) 2y_1 y_2 + (-2x) y_1^2 = 0$ Divide by $2y_1$, we get (01) $(1-x^2) y_2 - xy_1 = 0$ .....(1) Differentiate equation (1) n times w.r.to x by using Leibnitz's rule i.e. $D^n (u \cdot v) = u_n v + n\, u_{n-1} v_1 + \frac{n(n-1)}{2!} u_{n-2} v_2 + ...$ $.... + v_n u = 0$, we get $D^n [(1-x^2) y_2] - D^n [xy_1] = 0$ $\Rightarrow (1-x^2)(y_2)_n + n(y_2)_{n-1}(1-x^2)_1$ $+ \frac{n(n-1)}{2!}(y_2)_{n-2}(1-x^2)_2$ (01) $-[x(y_1)_n + n(y_1)_{n-1}(x)_1] = 0$ $\Rightarrow \left[(1-x^2) y_{n+2} + n\, y_{n+1}(-2x) + \frac{n(n-1)}{2!} y_n (-2)\right]$ $-[x\, y_{n+1} + n\, y_n] = 0$ $\Rightarrow (1-x^2) y_{n+2} - (2n+1) xy_{n+1} - (n^2 - n + n) y_n = 0$ $\therefore \boxed{(1-x^2) y_{n+2} - (2n+1)x\, y_{n+1} - n^2 y_n = 0}$ Ans. (01) which is the required result.		5				
7.	Unit 1	If prove that	10		Medium	Easy	CO1	Descriptive

		$y_1 = \frac{1}{1+x^2}$ $\Rightarrow (1+x^2)\, y_1 = 1$ Again differentiating w.r.to x, we get $(1+x^2)\, y_2 + 2xy_1 = 0$ Differentiate above equation n times w.r.to x, using Leibnitz's rule i.e. $D^n(uv) = u_n v + nu_{n-1} v + \frac{n(n-1)}{2!} u_{n-2} v_2 + \ldots + v_n u,$ we get $D^n[(1+x^2)\, y_2] + 2D^n[xy_1] = 0$ $\Rightarrow \left[(1+x^2)\,(y_2)_n + n(y_2)_{n-1}(1+x^2)_1 + \frac{n(n-1)}{2!}(y_2)_{n-2}(1+x^2)_2\right]$ $+ 2\left[x(y_1)_n + n\,(y_1)_{n-1}(x)_1\right] = 0$ $\Rightarrow (1+x^2)\, y_{n+2} + 2xny_{n+1} + \frac{n(n-1)}{2} y_n \cdot 2$ $+2xy_{n+1} + 2n\, y_n = 0$ $\Rightarrow (1+x^2)\, y_{n+2} + 2(n+1)x\, y_{n+1} + (n^2 - n + 2n)\, y_n = 0$ $\therefore \boxed{(1+x^2)\, y_{n+2} + 2(n+1)\, xy_{n+1} + n(n+1)\, y_n = 0}$ **Ans.** which is the required result. Scanned with CamScanner		5				
8.	Unit 1	If show that	10		Medium	Easy	CO1	Descriptive

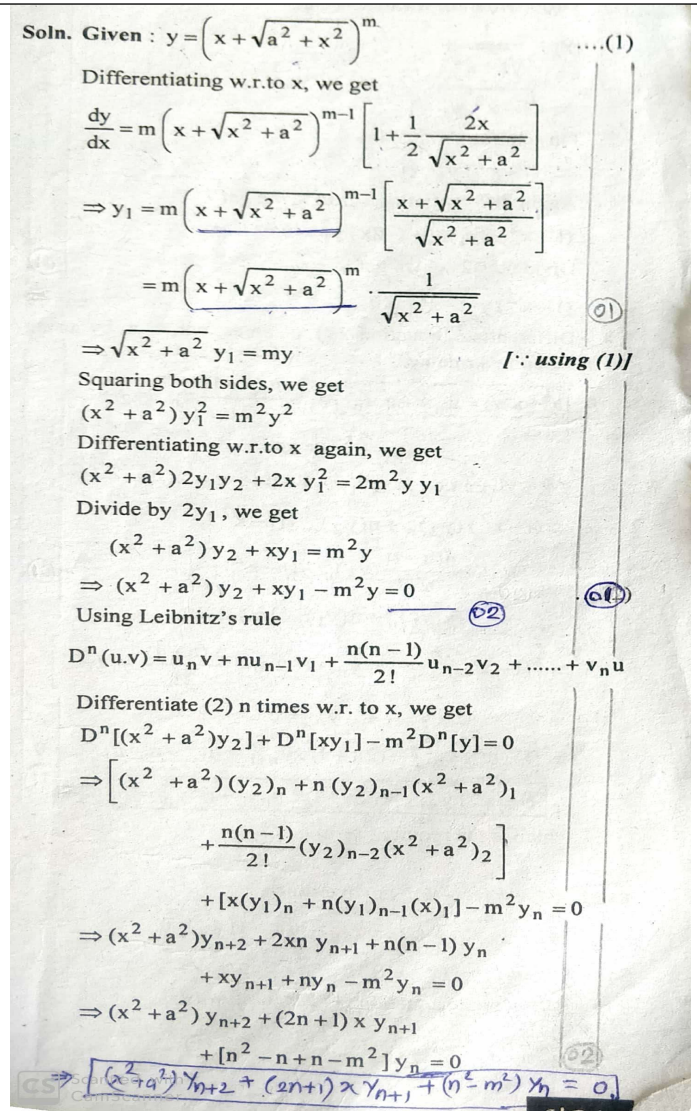
		Soln. Given: $y = e^{x^2/2}\cos x$ ....(1) Differentiating equation (1) w.r.to x, we get $y_1 = e^{x^2/2}\frac{1}{2}(2x)\cos x - e^{x^2/2}\sin x$ $\therefore\ y_1 = xy - e^{x^2/2}\sin x$ ....(2) *($\because\ y = e^{x^2/2}\cos x$)* Again differentiating w.r.to x, we get $y_2 = xy_1 + y - xe^{x^2/2}\sin x - e^{x^2/2}\cos x$ $\Rightarrow y_2 = xy_1 + y - xe^{x^2/2}\sin x - y$ $= xy_1 - x(xy - y_1)$ *($\because$ by using (2))* $= xy_1 - x^2y + xy_1$ $= 2xy_1 - x^2y$ $\Rightarrow y_2 - 2xy_1 + x^2y = 0$ ....(3) Using Leibnitz's rule, differentiate (3) 2n times w. r. to x, we get $D^{2n}(y_2) - 2D^{2n}(xy_1) + D^{2n}(x^2y) = 0$ $\Rightarrow y_{2n+2} - 2\left[x(y_1)_{2n} + 2n(y_1)_{2n-1}(x)_1 + 0\right]$ $+\left[x^2(y)_{2n} + 2n(y)_{2n-1}(x^2)_1 + \frac{2n(2n-1)}{2!}(y)_{2n-2}(x^2)_2\right] = 0$ $\Rightarrow y_{2n+2} - 2[xy_{2n+1} + 2n\,y_{2n}] + [x^2y_{2n} + 4nx\,y_{2n-1} + 2n(2n-1)y_{2n-2}] = 0$ $\Rightarrow y_{2n+2} - 2x\,y_{2n+1} - 4ny_{2n} + x^2y_{2n} + 4nx\,y_{2n-1} + 2n(2n-1)y_{2n-2} = 0$ .... (4) Put $x = 0$ in equation (4), we get $y_{2n+2}(0) + 0 - 4n\,y_{2n}(0) + 0 + 0 + 2n(2n-1)y_{2n-2}(0) = 0$ Scanned with CamScanner $\therefore\ \boxed{y_{2n+2}(0) - 4n\,y_{2n}(0) + 2n(2n-1)y_{2n-2}(0) = 0}$ **Ans.** which is required result. Scanned with CamScanner		5				
9.	Unit 1	Find the derivative of : , (	12		Medium	Moderate	CO1	Descriptive

		Soln. Given : $y = x^2 e^x$, which is a product of two functions. $\therefore$ Using Leibnitz's rule $D^n(u \cdot v) = u_n v + n u_{n-1} v_1 + \frac{n(n-1)}{2!} u_{n-2} v_2 + \ldots\ldots + v_n u.$ and taking $u = e^x$ & $v = x^2$, we get $D^n(x^2 e^x) = (e^x)_n x^2 + n(e^x)_{n-1}(x^2)_1 + \frac{n(n-1)}{2!}(e^x)_{n-2}(x^2)_2 + 0$ [$\because$ *higher derivatives are zero*] $= e^x x^2 + n e^x (2x) + \frac{n(n-1)}{2!} e^x (2)$ [$\because D^n(e^x) = e^x$] $\therefore y_n = x^2 e^x + 2xn e^x + (n^2 - n) e^x$ $\therefore \boxed{\frac{d^n y}{dx^n} = e^x [x^2 + 2xn + n^2 - n]}$ **Ans.** $\Rightarrow (1 + x^2) y_1 = 1$ Differentiating n times by using Leibnitz's rule $D^n(u.v) = u_n v + n u_{n-1} v_1 + \frac{n(n-1)}{2} u_{n-2} v_2 + \ldots\ldots + v_n u$, we get $(1 + x^2) y_{n+1} + n\, 2x y_n + \frac{n(n-1)}{2} 2 y_{n-1} = 0$ $\therefore \boxed{(1 + x^2) y_{n+1} + 2nx y_n + n(n-1) y_{n-1} = 0}$ **Ans.** which is the required result.		5				
10.	Unit 1	Using Taylor's series find the value of correct to four decimal places.	12		Medium	Moderate	CO1	Numerical

		Soln. By Taylor's series, we have $f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \ldots\ldots$ ...(1) Now, $64^\circ = 60^\circ + 4^\circ$ $\cos(64^\circ) = \cos(60^\circ + 4^\circ)$ Consider $x = 60^\circ$ & $h = 4^\circ = \frac{\pi}{180} \times 4 = \frac{\pi}{45} = 0.069813$ radian $\therefore (1) \Rightarrow \cos(64^\circ) = f(60^\circ) + hf'(60^\circ) + \frac{h^2}{2!} f''(60^\circ) + \frac{h^3}{3!} f'''(60^\circ) + \ldots\ldots$ ... (2) Here, $f(x) = \cos x$, $\therefore f(60^\circ) = \cos 60^\circ = \frac{1}{2}$ Differentiating successively w.r.to x, $\Rightarrow f'(x) = -\sin x$, $\therefore f'(60^\circ) = -\sin 60^\circ = -\frac{\sqrt{3}}{2}$ $\Rightarrow f''(x) = -\cos x$, $\therefore f''(60^\circ) = -\cos 60^\circ = -\frac{1}{2}$ $\Rightarrow f'''(x) = \sin x$, $\therefore f'''(60^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2}$ $\therefore$ Equation (2) $\Rightarrow$ $\cos(60^\circ + h) = \cos(64^\circ)$ $= \frac{1}{2} - \frac{\sqrt{3}}{2} h + \frac{h^2}{2}\left(-\frac{1}{2}\right) + \frac{h^3}{3!}\left(\frac{\sqrt{3}}{2}\right) + \ldots$ $= \frac{1}{2} - \frac{\sqrt{3}}{2} h - \frac{h^2}{4} + \frac{\sqrt{3}\, h^3}{2\,(6)} + \ldots\ldots$ $\therefore \cos 64^\circ = \frac{1}{2} - \frac{\sqrt{3}}{2}(0.069813) - \frac{(0.069813)^2}{4} + \frac{\sqrt{3}}{2}\frac{(0.069813)^3}{6} + \ldots\ldots$ *[$\because$ $h = 0.069813$ radian]* $= \frac{1}{2} - 0.06045 - 0.00121846 + 0.000049112$ $\therefore \boxed{\cos 64^\circ = 0.4383}$ **Ans.**		5				
11.	Unit 1	Given find by using Taylor's theorem.	12		Medium	Moderate	CO1	Numerical

		Soln. We have by Taylor's theorem $f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \ldots\ldots$ … (1) Consider $f\left(\frac{11}{10}\right) = f\left(1 + \frac{1}{10}\right) = f(1 + 0.1)$ Taking $x = 1$ & $h = 0.1$ $\therefore$ Equation (1) $\Rightarrow f\left(\frac{11}{10}\right) = f(1) + (0.1)f'(1) + \frac{(0.1)^2}{2!}f''(1) + \frac{(0.1)^3}{3!}f'''(1) + \ldots\ldots$ ….(2) **Given:** $f(x) = x^3 + 8x^2 + 15x - 24$ $\therefore\ f(1) = 1 + 8 + 15 - 24 = 0$ Also, $f'(x) = 3x^2 + 16x + 15$ $\therefore\ f'(1) = 3 + 16 + 15 = 34$ $f''(x) = 6x + 16$ $\therefore\ f''(1) = 6 + 16 = 22$ and $f'''(x) = 6$ $\therefore\ f'''(1) = 6$ $\therefore$ Equation (2) $\Rightarrow$ $f\left(\frac{11}{10}\right) = 0 + 0.1(34) + \frac{0.01}{2}(22) + \frac{0.001}{6}6 = 3.511$ $\therefore\ \boxed{f\left(\frac{11}{10}\right) = 3.511}$ **Ans.** Scanned with CamScanner		5				
12.	Unit 1	Using Taylor's series find the value of correct up to five decimal places.	12		Medium	Moderate	CO1	Numerical

		Soln. Taylor's series expansion is $f(x+h) = f(x) + \frac{h}{1!}f'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \dots$ Consider $f(x) = \sin x$ $\therefore \sin(x+h) = \sin x + h\cos x + \frac{h^2}{2!}(-\sin x) + \frac{h^3}{3!}(-\cos x) + \dots$ Let $x = 60^\circ$, $h = 30' = \frac{1^\circ}{2} = 0.008725$ radian $\therefore x + h = 60^\circ + 0.008725$ $\Rightarrow \sin(60^\circ 30') = \sin 60^\circ + 0.008725 \cos 60^\circ - \frac{(0.008725)^2}{2!}\sin 60^\circ - \frac{(0.008725)^3}{3!}\cos 60^\circ + \dots\dots$ $= 0.86603 + (0.008725)(0.5) - \frac{(0.008725)^2}{2!}(0.86603) - \frac{(0.008725)^3}{3!}(0.5) + \dots\dots$ $= 0.86603 + 0.0043625 - 0.00003296 - 0.000000055$ $\therefore \boxed{\sin 60^\circ 30' = 0.87036}$ **Ans.**		5				
13.	Unit 1	If, prove that	12		Medium	Moderate	CO1	Descriptive

				5				
14.	Unit 1	Prove that:	12		Medium	Moderate	CO1	Descriptive

		Soln. Let $f(x) = a^x \Rightarrow f(0) = a^0 = 1$ $\therefore f'(x) = a^x \log a \Rightarrow f'(0) = a^0 \log a = \log a$ $\therefore f''(x) = a^x (\log a)^2 \Rightarrow f''(0) = a^0 (\log a)^2 = (\log a)^2$ $\therefore f'''(x) = a^x (\log a)^3 \Rightarrow f'''(0) = a^0 (\log a)^3 = (\log a)^3$ ⋮ & so on. Now, the Maclaurin's series is given by $f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \ldots\ldots$ $\therefore \boxed{a^x = 1 + x \log a + \frac{x^2}{2!} (\log a)^2 + \frac{x^3}{3!} (\log a)^3 + \ldots\ldots}$ **Ans.** which is the required result.		5				
15.	Unit 1	Expand in ascending powers of upto and including the term and calculate up to three places.	12		Medium	Moderate	CO1	Numerical

Let $y = \log(\cos x)$ $y(0)=0$

Sol^{n} :- Differentiating successively w.r.to x,

$$\Rightarrow y_1 = \frac{-\sin x}{\cos x} = -\tan x \quad \therefore \; y_1(0) = 0$$

$$\Rightarrow y_2 = -\sec^2 x \quad \therefore \; y_2(0) = -1$$

$$\Rightarrow y_3 = -2\sec x \,.\, \sec x \tan x$$

$$\Rightarrow y_3 = -2\sec^2 x \,.\, \tan x \quad \therefore y_3(0) = 0$$

$$\Rightarrow y_4 = -2\left[\sec^2 x \sec^2 x + \tan x \,.\, 2\sec x \,.\, \sec x \tan x\right]$$

$$\therefore \; y_4(0) = -2$$

$\therefore$ By Maclaurin's series, we have

$$y(x) = y(0) + xy_1(0) + \frac{x^2}{2!}y_2(0) + \frac{x^3}{3!}y_3(0) + \frac{x^4}{4!}y_4(0) + \ldots$$

$$\therefore \log\cos x = \frac{x^2}{2}(-1) + \frac{x^4}{4!}(-2)$$

$$\therefore \log(\cos x) = -\frac{x^2}{2} - \frac{x^4}{12} + \ldots\ldots$$

$$\Rightarrow \log\left(\cos\frac{\pi}{12}\right) = -\frac{\pi^2}{288} - \frac{\pi^4}{248332}$$

$$= -0.0343 - 0.000391 = -0.03469$$

$$\therefore \log_{10}\left(\cos\frac{\pi}{12}\right) = \frac{\log_e\left(\cos\frac{\pi}{12}\right)}{\log_e 10} = \frac{-0.03469}{2.3025}$$

$$\therefore \log_{10}\left(\cos\frac{\pi}{12}\right) = -0.015066$$ **Ans.**

Scanned with CamScanner

5

16. Unit 1 If prove that 12 Medium Moderate CO1 Descriptive

Soln. Given : $y = e^{m\cos^{-1}x}$ ....(1)

Differentiating (1) w. r. to x, we get

$$y_1 = e^{m\cos^{-1}x}\left(\frac{-m}{\sqrt{1-x^2}}\right)$$

$$\Rightarrow \sqrt{1-x^2}\, y_1 = -me^{m\cos^{-1}x}$$

$$\Rightarrow \sqrt{1-x^2}\, y_1 = -my \quad \text{...(2)}$$

Squaring both sides, we get

$$(1-x^2)y_1^2 = m^2y^2$$

Again differentiating w.r.to x, we get

$$(1-x^2)2y_1y_2 - 2xy_1^2 = 2m^2y\, y_1$$

$$\Rightarrow (1-x^2)y_2 - xy_1 = m^2y$$

$$\Rightarrow (1-x^2)y_2 - xy_1 - m^2y = 0 \quad \text{....(3)}$$

Now, differentiating n times w.r.to x using Leibnitz's rule, we get

$$D^n[(1-x^2)\, y_2] - D^n[xy_1] - m^2\, D^n[y] = 0$$

CS Scanned with CamScanner

$$\Rightarrow \left[(1-x^2)(y_2)_n + n(y_2)_{n-1}(1-x^2)_1 + \frac{n(n-1)}{2!}(y_2)_{n-2}(1-x^2)_2\right]$$
$$-[x(y_1)_n + n(y_1)_{n-1}(x)_1] - m^2 y_n = 0$$
$$\Rightarrow \left[(1-x^2)y_{n+2} - 2nxy_{n+1} + \frac{n(n-1)}{2}y_n \cdot (-2)\right]$$
$$-[xy_{n+1} + ny_n] - m^2 y_n = 0$$
$$\Rightarrow (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 - n + n + m^2)y_n = 0$$
$$\therefore \quad (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (m^2+n^2)y_n = 0 \qquad \ldots\ldots(4)$$

put $x = 0$ in (1), (2), (3) & (4), we get

$(y)_0 = e^{m\cos^{-1}0} = e^{m\pi/2}$, $\quad (\because y_0 = e^{m\pi/2})$

$(y_1)_0 = -m(y)_0 = -me^{m\pi/2}$, $\quad (\because y_0 = e^{m\pi/2})$

$(y_2)_0 = m^2(y)_0 = m^2 e^{m\pi/2}$,

and $(y_{n+2})_0 = (n^2+m^2)(y_n)_0 \qquad \ldots\ldots(5)$

Case I : Putting $n = 1, 3, 5, \ldots\ldots$ in equation (5), we get

For $n = 1$, $(y_3)_0 = (1^2+m^2)(y_1)_0$
$= -m(1^2+m^2)e^{m\pi/2}$

For $n = 3$, $(y_5)_0 = (3^2+m^2)(y_3)_0$
$= -m(1^2+m^2)(3^2+m^2)e^{m\pi/2}$

For $n = 5$, $(y_7)_0 = (5^2+m^2)(y_5)_0$
$= -m(1^2+m^2)(3^2+m^2)(5^2+m^2)e^{m\pi/2}$

$\vdots$

So in general,

$$(y_n)_0 = [(n-2)^2+m^2]\ldots\ldots \ldots\ldots (5^2+m^2)(3^2+m^2)(1^2+m^2)(-me^{m\pi/2})$$

Ans.

when n is odd.

Case II : Putting $n = 2, 4, 6, \ldots\ldots$ in equation (5), we get

For $n = 2$, $(y_4)_0 = (2^2+m^2)(y_2)_0$
$= m^2(2^2+m^2)e^{m\pi/2}$

For $n = 4$, $(y_6)_0 = (4^2+m^2)(y_4)_0$
$= m^2(2^2+m^2)(4^2+m^2)e^{m\pi/2}$

For $n = 6$, $(y_8)_0 = (6^2+m^2)(y_6)_0$
$= m^2(2^2+m^2)(4^2+m^2)(6^2+m^2)e^{m\pi/2}$

$\vdots$

Scanned with CamScanner

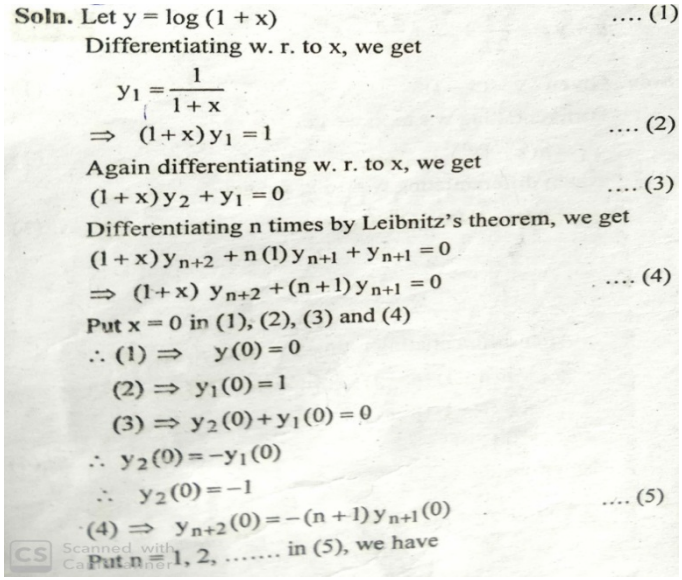
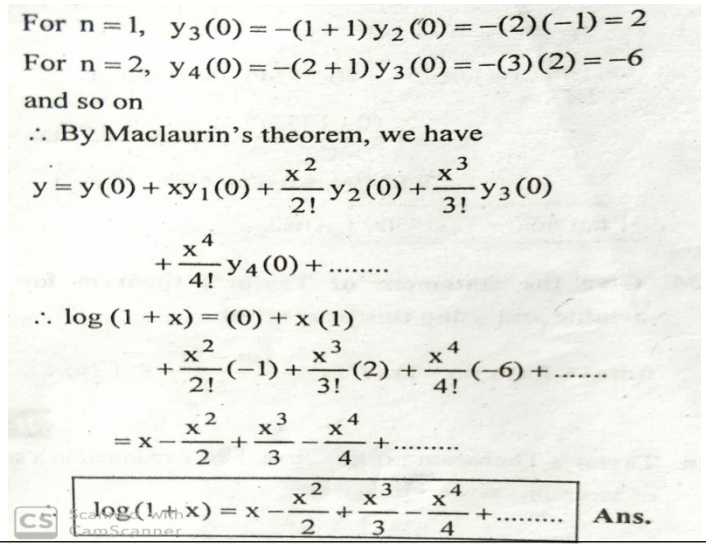
So in general,

$$(y_n)_0 = m^2[(n-2)^2+m^2]\ldots\ldots \ldots\ldots (6^2+m^2)(4^2+m^2)(2^2+m^2)e^{m\pi/2}$$

Ans.

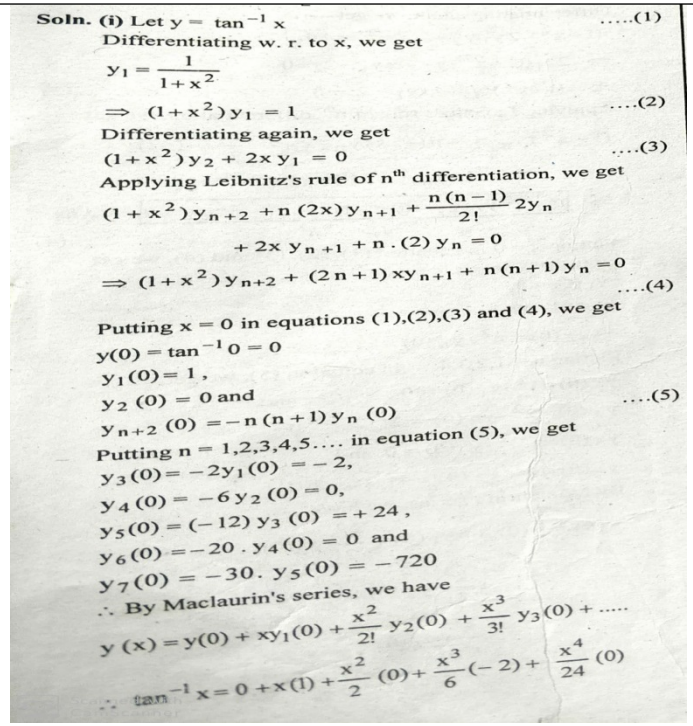
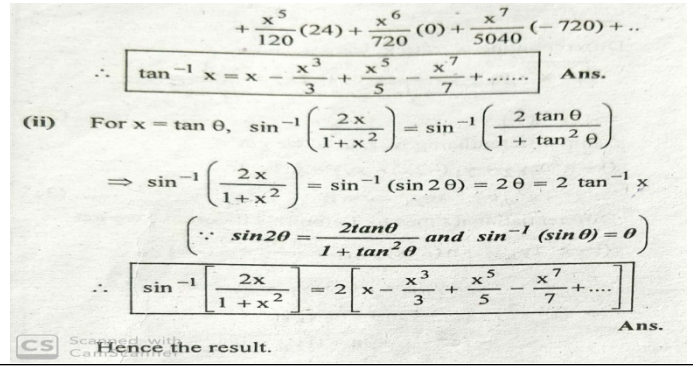
when n is even.

Scanned with CamScanner

	Unit 1	Use Maclaurin's theorem to show that  **Soln.** Let $y = \log(1+x)$ .... (1) Differentiating w. r. to x, we get $y_1 = \frac{1}{1+x}$ $\Rightarrow (1+x)y_1 = 1$ .... (2) Again differentiating w. r. to x, we get $(1+x)y_2 + y_1 = 0$ .... (3) Differentiating n times by Leibnitz's theorem, we get $(1+x)y_{n+2} + n(1)y_{n+1} + y_{n+1} = 0$ $\Rightarrow (1+x)\ y_{n+2} + (n+1)y_{n+1} = 0$ .... (4) Put x = 0 in (1), (2), (3) and (4) $\therefore$ (1) $\Rightarrow$ $y(0) = 0$ (2) $\Rightarrow$ $y_1(0) = 1$ (3) $\Rightarrow$ $y_2(0) + y_1(0) = 0$ $\therefore\ y_2(0) = -y_1(0)$ $\therefore\ y_2(0) = -1$ (4) $\Rightarrow$ $y_{n+2}(0) = -(n+1)y_{n+1}(0)$ .... (5) Put n = 1, 2, ... in (5), we have  For $n = 1$, $y_3(0) = -(1+1)y_2(0) = -(2)(-1) = 2$ For $n = 2$, $y_4(0) = -(2+1)y_3(0) = -(3)(2) = -6$ and so on $\therefore$ By Maclaurin's theorem, we have $y = y(0) + xy_1(0) + \frac{x^2}{2!}y_2(0) + \frac{x^3}{3!}y_3(0) + \frac{x^4}{4!}y_4(0) + ...$ $\therefore \log(1+x) = (0) + x(1) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(2) + \frac{x^4}{4!}(-6) + ...$ $= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + ...$ $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + ...$ **Ans.**	12	5	Medium	Moderate	CO1	Descriptive
18.	Unit 1	Use Maclaurin's theorem to show that	12		Medium	Moderate	CO1	Descriptive

		Soln. Let $y = \log(1+e^x)$ $\Rightarrow y(0) = \log 2$ Differentiating w.r.to x, we get $y_1 = \frac{e^x}{1+e^x}$ $\Rightarrow y_1(0) = \frac{1}{2}$ Again differentiating, we get $y_2 = \frac{(1+e^x)e^x - e^x.e^x}{(1+e^x)^2} = \frac{e^x}{(1+e^x)^2}$, $\Rightarrow y_2(0) = \frac{1}{4}$ Now, $y_3 = \frac{(1+e^x)^2 e^x - e^x\, 2(1+e^x)e^x}{(1+e^x)^4}$ $= \frac{(1+e^x)e^x - 2e^x e^x}{(1+e^x)^3}$ $= \frac{e^x - e^x.e^x}{(1+e^x)^3} = \frac{e^x - e^{2x}}{(1+e^x)^3}$ $\therefore y_3(0) = 0$ Also, $y_4 = \frac{(1+e^x)^3(e^x - 2e^{2x}) - (e^x - e^{2x})\,3e^x(1+e^x)^2}{(1+e^x)^6}$ $= \frac{(1+e^x)(e^x - 2e^{2x}) - 3e^x(e^x - e^{2x})}{(1+e^x)^4}$ $= \frac{e^x - 2e^{2x} + e^{2x} - 2e^{3x} - 3e^{2x} + 3e^{3x}}{(1+e^x)^4}$ $= \frac{e^x - 4e^{2x} + e^{3x}}{(1+e^x)^4}$ $\therefore y_4(0) = \frac{-2}{2^4} = \frac{-1}{8}$ By Maclaurin's theorem, we have $y(x) = y(0) + xy_1(0) + \frac{x^2}{2!}y_2(0) + \frac{x^3}{3!}y_3(0) + \frac{x^4}{4!}y_4(0) + \ldots$ $\therefore \log(1+e^x) = \log 2 + x\left(\frac{1}{2}\right) + \frac{x^2}{2}\left(\frac{1}{4}\right) + \frac{x^3}{6}(0) + \frac{x^4}{24}\left(\frac{-1}{8}\right) + \ldots$ $\therefore \boxed{\log(1+e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{1}{192}x^4 + \ldots}$ **Ans.** which is the required result.		5				
19.	Unit 1	Obtain the Maclaurin's expansion of and hence find the value of correct up to four decimal places.	12		Medium	Difficult	CO1	Numerical

		Ans. Let $f(x)=\tan\left(\frac{\pi}{4}+x\right)$ $\therefore\ f(0)=\tan\frac{\pi}{4}=1$ Now $f'(x)=\sec^2\left(\frac{\pi}{4}+x\right)=1+\tan^2\left(\frac{\pi}{4}+x\right)$ $=1+[f(x)]^2$ $\therefore\ f'(0)=1+[f(0)]^2$ $=1+1$ $=2$ Thus $f''(x)=2f(x)f'(x)$ $\therefore\ f''(0)=2f(0)f'(0)$ $=2\times1\times2=4$ And $f'''(x)=2f'(x)f'(x)+2f(x)f''(x)$ $=2[f'(x)]^2+2f(x)f''(x)$ $\therefore\ f'''(0)=2[f'(0)]^2+2f(0)f''(0)$ $=2(2)^2+2(1)(4)$ $=8+8$ $=16$ and so on. By Maclaurin's series, we have $f(x)=f(0)+xf'(0)+\frac{x^2}{2!}f''(0)+\frac{x^3}{3!}f'''(0)+\ldots..$ $\therefore\ \boxed{\tan\left(\frac{\pi}{4}+x\right)=1+2x+2x^2+\frac{8}{3}x^3+\ldots.}$ **Ans.** ..(1) Now $46^\circ30'=45^\circ+1^\circ30'=45^\circ+90'$ $=\frac{\pi}{4}+\frac{\pi}{120}=\left(\frac{\pi}{4}+0.02618 \text{ radian}\right)$ Putting $x=1^\circ30'=0.02618$ in equation (1), we get $\tan 46^\circ30'=1+2(0.02618)+2(0.02618)^2+\frac{8}{3}(0.02618)^3+\ldots.$ $=1+0.05236+0.00137+0.00005+\ldots.$ $\boxed{\tan 46^\circ30'=1.0538 \text{ approximately}}$ **Ans.**		5				
20.	Unit 1	Obtain the expansion of in ascending powers and hence show that:	12		Medium	Difficult	CO1	Descriptive

		 		5				
21.	Unit 1	If show that Hence show that .	15		High	Very Difficult	CO1	Descriptive

Soln. **(i)** Let $y = \tan^{-1} x$ .....(1)

Differentiating w. r. to x, we get

$$y_1 = \frac{1}{1+x^2}$$

$$\Rightarrow (1+x^2) y_1 = 1 \quad \text{....(2)}$$

Differentiating again, we get

$$(1+x^2) y_2 + 2x y_1 = 0 \quad \text{....(3)}$$

Applying Leibnitz's rule of n^{th} differentiation, we get

$$(1+x^2) y_{n+2} + n(2x) y_{n+1} + \frac{n(n-1)}{2!} 2 y_n + 2x y_{n+1} + n.(2) y_n = 0$$

$$\Rightarrow (1+x^2) y_{n+2} + (2n+1) x y_{n+1} + n(n+1) y_n = 0 \quad \text{....(4)}$$

Putting $x = 0$ in equations (1),(2),(3) and (4), we get

$y(0) = \tan^{-1} 0 = 0$

$y_1(0) = 1$,

$y_2(0) = 0$ and

$y_{n+2}(0) = -n(n+1) y_n(0)$ ....(5)

Putting $n = 1,2,3,4,5....$ in equation (5), we get

$y_3(0) = -2y_1(0) = -2$,

$y_4(0) = -6 y_2(0) = 0$,

$y_5(0) = (-12) y_3(0) = +24$,

$y_6(0) = -20 . y_4(0) = 0$ and

$y_7(0) = -30 . y_5(0) = -720$

$\therefore$ By Maclaurin's series, we have

$$y(x) = y(0) + x y_1(0) + \frac{x^2}{2!} y_2(0) + \frac{x^3}{3!} y_3(0) + ...$$

$$\therefore \tan^{-1} x = 0 + x(1) + \frac{x^2}{2}(0) + \frac{x^3}{6}(-2) + \frac{x^4}{24}(0) + \frac{x^5}{120}(24) + \frac{x^6}{720}(0) + \frac{x^7}{5040}(-720) + ..$$

$$\therefore \boxed{\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + ...} \quad \textbf{Ans.}$$

(ii) For $x = \tan\theta$, $\sin^{-1}\left(\frac{2x}{1+x^2}\right) = \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$

$$\Rightarrow \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \sin^{-1}(\sin 2\theta) = 2\theta = 2\tan^{-1} x$$

$$\left(\because \sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta} \text{ and } \sin^{-1}(\sin\theta) = \theta\right)$$

$$\therefore \boxed{\sin^{-1}\left[\frac{2x}{1+x^2}\right] = 2\left[x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + ...\right]}$$

Ans.

Hence the result.

Soln. Given: $y = \sin(m \sin^{-1} x)$ ....(1)

Differentiating w.r.to x, we get

$$y_1 = \cos(m \sin^{-1} x)\left(\frac{m}{\sqrt{1-x^2}}\right)$$

$$\Rightarrow \sqrt{1-x^2}\, y_1 = m \cos(m \sin^{-1} x) \quad \text{....(2)}$$

Squaring, we get

$$(1-x^2) y_1^2 = m^2 \cos^2(m \sin^{-1} x)$$

$$= m^2 [1 - \sin^2(m \sin^{-1} x)]$$

$$\Rightarrow (1-x^2) y_1^2 = m^2 (1-y^2) \quad (\because \textit{using (1)})$$

Again differentiating w.r.to x, we get

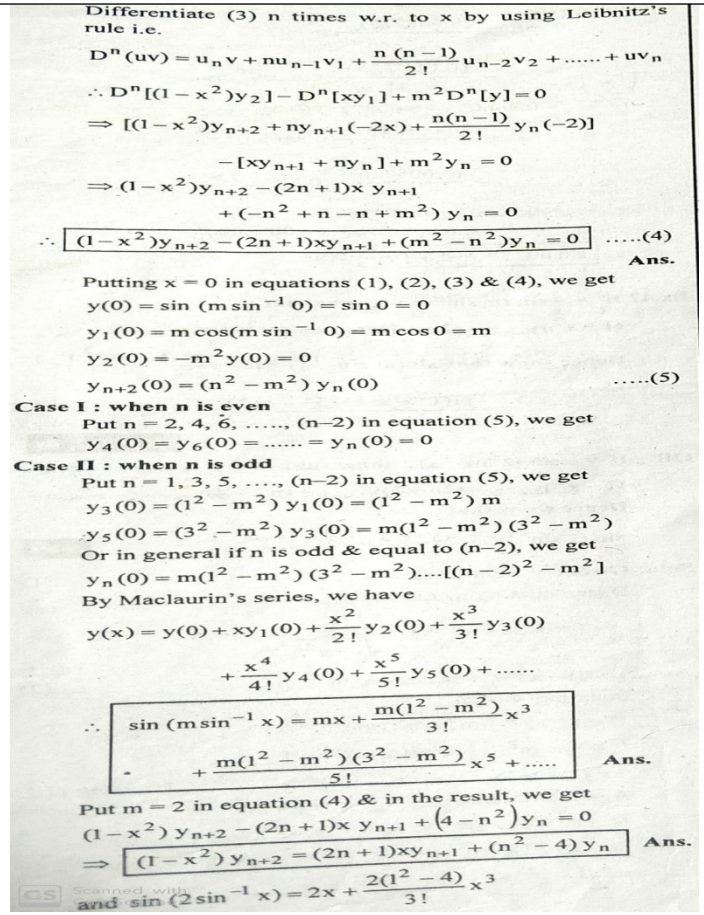
$$(1-x^2) 2y_1 y_2 - 2x y_1^2 = m^2(-2y\, y_1)$$

$$\Rightarrow 2y_1 [(1-x^2) y_2 - x y_1 + m^2 y] = 0$$

$$\Rightarrow (1-x^2) y_2 - x y_1 + m^2 y = 0 \quad \text{.....(3)}$$

Scanned with CamScanner

5

								
22.	Unit 1	If, prove that	15		High	Very Difficult	CO1	Descriptive

Differentiate (3) n times w.r. to x by using Leibnitz's rule i.e.

$$D^n(uv) = u_n v + n u_{n-1} v_1 + \frac{n(n-1)}{2!} u_{n-2} v_2 + \ldots\ldots + u v_n$$

$$\therefore D^n[(1-x^2)y_2] - D^n[xy_1] + m^2 D^n[y] = 0$$

$$\Rightarrow [(1-x^2)y_{n+2} + n y_{n+1}(-2x) + \frac{n(n-1)}{2!} y_n(-2)]$$

$$- [xy_{n+1} + n y_n] + m^2 y_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - (2n+1)x\, y_{n+1} + (-n^2 + n - n + m^2)\, y_n = 0$$

$$\therefore \boxed{(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0} \quad \ldots\ldots(4)$$

Ans.

Putting $x = 0$ in equations (1), (2), (3) & (4), we get

$$y(0) = \sin(m \sin^{-1} 0) = \sin 0 = 0$$

$$y_1(0) = m\cos(m \sin^{-1} 0) = m \cos 0 = m$$

$$y_2(0) = -m^2 y(0) = 0$$

$$y_{n+2}(0) = (n^2 - m^2)\, y_n(0) \quad \ldots\ldots(5)$$

Case I : when n is even

Put $n = 2, 4, 6, \ldots\ldots, (n-2)$ in equation (5), we get

$$y_4(0) = y_6(0) = \ldots\ldots = y_n(0) = 0$$

Case II : when n is odd

Put $n = 1, 3, 5, \ldots\ldots, (n-2)$ in equation (5), we get

$$y_3(0) = (1^2 - m^2)\, y_1(0) = (1^2 - m^2)\, m$$

$$y_5(0) = (3^2 - m^2)\, y_3(0) = m(1^2 - m^2)(3^2 - m^2)$$

Or in general if n is odd & equal to (n–2), we get

$$y_n(0) = m(1^2 - m^2)(3^2 - m^2)\ldots[(n-2)^2 - m^2]$$

By Maclaurin's series, we have

$$y(x) = y(0) + xy_1(0) + \frac{x^2}{2!} y_2(0) + \frac{x^3}{3!} y_3(0)$$

$$+ \frac{x^4}{4!} y_4(0) + \frac{x^5}{5!} y_5(0) + \ldots\ldots$$

$$\therefore \boxed{\sin(m \sin^{-1} x) = mx + \frac{m(1^2 - m^2)}{3!} x^3 + \frac{m(1^2 - m^2)(3^2 - m^2)}{5!} x^5 + \ldots\ldots}$$

Ans.

Put $m = 2$ in equation (4) & in the result, we get

$$(1-x^2)\, y_{n+2} - (2n+1)x\, y_{n+1} + (4 - n^2) y_n = 0$$

$$\Rightarrow \boxed{(1-x^2)\, y_{n+2} = (2n+1)xy_{n+1} + (n^2 - 4)\, y_n}$$

Ans.

and $\sin(2 \sin^{-1} x) = 2x + \frac{2(1^2 - 4)}{3!} x^3$

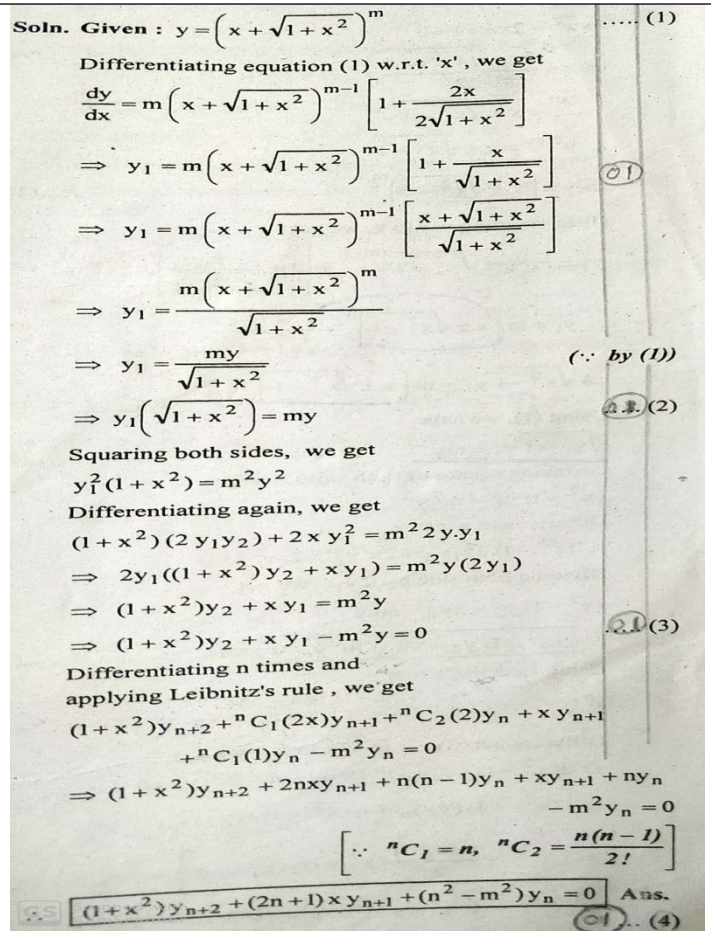
$$+ \frac{2(1^2 - 4)(3^2 - 4)}{5!} x^5 + \ldots\ldots$$

$$= 2x + \frac{(-3).2}{6} x^3 + \frac{2(-3).5}{120} x^5 + \ldots\ldots$$

$$\therefore \boxed{\sin(2 \sin^{-1} x) = 2x - x^3 - \frac{x^5}{4} + \ldots\ldots}$$

Ans.

which is the required result.

				5				

Soln. Given : $y=\left(x+\sqrt{1+x^2}\right)^m$ ..... (1)

Differentiating equation (1) w.r.t. 'x' , we get

$$\frac{dy}{dx}=m\left(x+\sqrt{1+x^2}\right)^{m-1}\left[1+\frac{2x}{2\sqrt{1+x^2}}\right]$$

$$\Rightarrow \quad y_1=m\left(x+\sqrt{1+x^2}\right)^{m-1}\left[1+\frac{x}{\sqrt{1+x^2}}\right]$$

$$\Rightarrow \quad y_1=m\left(x+\sqrt{1+x^2}\right)^{m-1}\left[\frac{x+\sqrt{1+x^2}}{\sqrt{1+x^2}}\right]$$

$$\Rightarrow \quad y_1=\frac{m\left(x+\sqrt{1+x^2}\right)^m}{\sqrt{1+x^2}}$$

$$\Rightarrow \quad y_1=\frac{my}{\sqrt{1+x^2}} \qquad (\because \text{ by (1)})$$

$$\Rightarrow \quad y_1\left(\sqrt{1+x^2}\right)=my \qquad \text{(2)}$$

Squaring both sides, we get

$$y_1^2(1+x^2)=m^2y^2$$

Differentiating again, we get

$$(1+x^2)(2y_1y_2)+2xy_1^2=m^2 2y.y_1$$

$$\Rightarrow \quad 2y_1((1+x^2)y_2+xy_1)=m^2y(2y_1)$$

$$\Rightarrow \quad (1+x^2)y_2+xy_1=m^2y$$

$$\Rightarrow \quad (1+x^2)y_2+xy_1-m^2y=0 \qquad \text{(3)}$$

Differentiating n times and applying Leibnitz's rule , we get

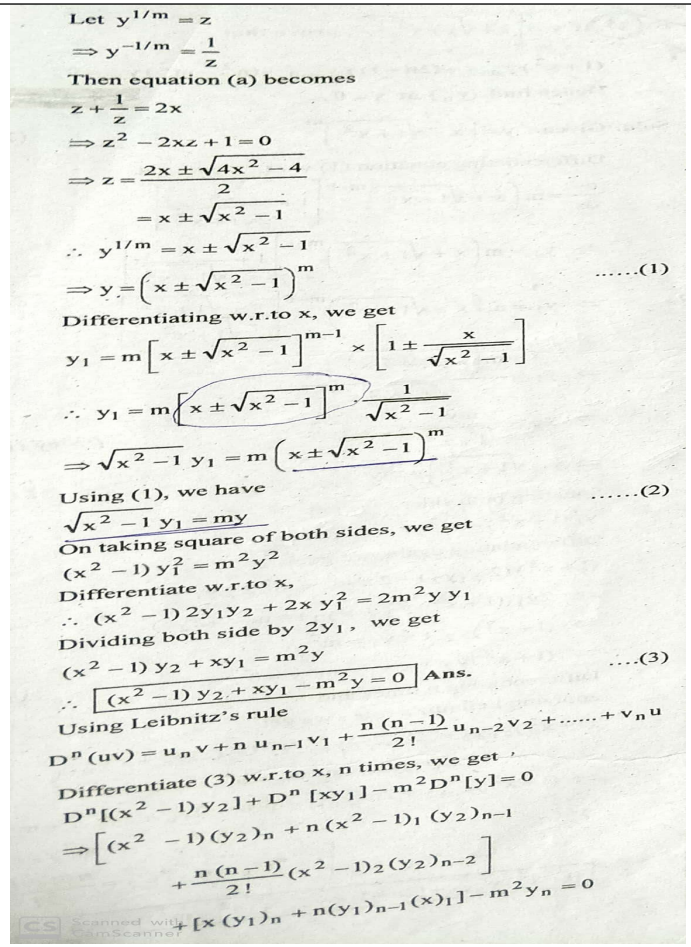
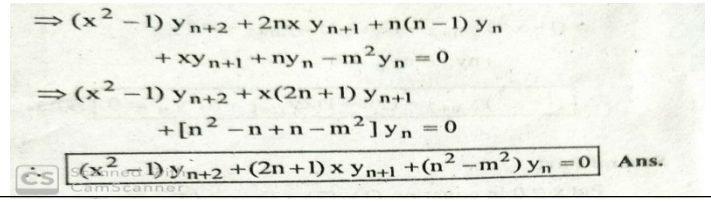
$$(1+x^2)y_{n+2}+{}^nC_1(2x)y_{n+1}+{}^nC_2(2)y_n+xy_{n+1}+{}^nC_1(1)y_n-m^2y_n=0$$

$$\Rightarrow (1+x^2)y_{n+2}+2nxy_{n+1}+n(n-1)y_n+xy_{n+1}+ny_n-m^2y_n=0$$

$$\left[\because \ {}^nC_1=n, \ {}^nC_2=\frac{n(n-1)}{2!}\right]$$

$$\therefore \quad \boxed{(1+x^2)y_{n+2}+(2n+1)xy_{n+1}+(n^2-m^2)y_n=0} \quad \text{Ans.} \qquad \text{(4)}$$

		Second part: Putting $x = 0$ in equation (1), (2), (3) & (4), we get $y(0) = (0+\sqrt{1+0})^m = (\sqrt{1})^m = 1^m = 1,$ $y_1(0) = my(0) = m,$ *[∵ y(0) = 1]* $y_2(0) = m^2 y(0) = m^2,$ *[∵ y(0) = 1]* and $y_{n+2}(0) = (m^2 - n^2)y_n(0)$ ..... (5) **If n is even,** putting $n = 2, 4, 6, \ldots, (n-2)$ in equation (5), we get For $n = 2$, $y_4(0) = (m^2 - 2^2)\, y_2(0) = (m^2 - 2^2)\, m^2$ *[∵ $y_2(0) = m^2$]* For $n = 4$, $y_6(0) = (m^2 - 4^2)\, y_4(0)$ $= (m^2 - 4^2)(m^2 - 2^2)\, m^2$ *[∵ $y_4(0) = (m^2 - 2^2)\, m^2$]* $\therefore \boxed{(y_n)_0 = [m^2 - (n-2)^2] \ldots\ldots, (m^2 - 2^2)m^2}$ **Ans.** **If n is odd,** putting $n = 1,3,5, \ldots\ldots (n-2)$ in equation (5), we get For $n = 1$, $y_3(0) = (m^2 - 1^2)\, y_1(0) = (m^2 - 1^2)\, m$ For $n = 3$ $y_5(0) = (m^2 - 3^2)\, y_3(0) = (m^2 - 3^2)(m^2 - 1^2)\, m$ $\therefore \boxed{(y_n)_0 = [m^2 - (n-2)^2] \ldots\ldots (m^2 - 1^2)m}$ **Ans.** Scanned with CamScanner						
23.	Unit 1	If, prove that	15		High	Very Difficult	CO1	Descriptive

		 		5				
24.	Unit 1	If prove that	15		High	Very Difficult	CO1	Descriptive

Soln. Given :

$$y = \left[\log\left\{x + \sqrt{1+x^2}\right\}\right]^2 \quad \ldots(1)$$

Differentiating w.r. to x, we get

$$y_1 = 2\log\left\{x + \sqrt{1+x^2}\right\} \cdot \frac{1}{x + \sqrt{1+x^2}} \left(1 + \frac{2x}{2\sqrt{1+x^2}}\right)$$

$$= 2\log\left\{x + \sqrt{1+x^2}\right\} \left(\frac{x + \sqrt{1+x^2}}{x + \sqrt{1+x^2}}\right) \cdot \frac{1}{\sqrt{1+x^2}}$$

$$\Rightarrow \sqrt{1+x^2}\, y_1 = 2\log\left\{x + \sqrt{1+x^2}\right\}$$

$$\Rightarrow \sqrt{1+x^2}\, y_1 = 2y^{1/2} \quad \ldots\ldots (2)$$

(∵ by using (1))

Squaring both sides, we get

$$(1+x^2)\, y_1^2 = 4y$$

Again differentiating w.r.to x, we get

$$(1+x^2)\, 2y_1 y_2 + 2xy_1^2 = 4y_1$$

Dividing by $2y_1$, we have

$$(1+x^2)\, y_2 + xy_1 = 2$$

$$\Rightarrow (1+x^2)\, y_2 + xy_1 - 2 = 0 \quad \ldots(3)$$

Differentiating equation (3) n times w.r.to x by using Leibnitz's rule, we have

$$D^n[1+x^2)y_2] + D^n[xy_1] - D^n[2] = 0$$

$$\Rightarrow \left[(1+x^2)\,(y_2)_n + n(y_2)_{n-1}(1+x^2)_1 + \frac{n(n-1)}{2!}(y_2)_{n-2}(1+x^2)_2\right]$$

$$+[x(y_1)_n + n(y_1)_{n-1}(x)_1] - 0 = 0$$

$$\Rightarrow (1+x^2)\, y_{n+2} + 2nx\, y_{n+1} + n(n-1)\, y_n + xy_{n+1} + ny_n = 0$$

$$\rightarrow (1+x^2)\, y_{n+2} + (2n+1)\, x\, y_{n+1} + n^2 y_n = 0$$

Scanned with

					5				